

# LSC Validation Infrastructure

## LSC Covariance And Data Policy

Status: Step 2 validation infrastructure

### 1. Purpose

This policy defines which data are essential, which assumptions are allowed, and which statistical shortcuts are forbidden before BEST-2-era validation.

### 2. Data Classification

| Class                 | Meaning                                                        |
|-----------------------|----------------------------------------------------------------|
| ESSENTIAL             | Required for likelihood-level validation or core falsification |
| IMPORTANT             | Strongly needed for reliable interpretation                    |
| OPTIONAL              | Useful but not required for the main validation decision       |
| CURRENTLY UNAVAILABLE | Not present in the current public/local package                |

### 3. Required Covariance Structures

| Data Object                 | Class     | Required Content                                        | Current Status        |
|-----------------------------|-----------|---------------------------------------------------------|-----------------------|
| BEST statistical covariance | ESSENTIAL | inner/outer/runs statistical correlations               | CURRENTLY UNAVAILABLE |
| BEST systematic covariance  | ESSENTIAL | source, extraction, efficiency, background correlations | CURRENTLY UNAVAILABLE |
| GALLEX covariance           | IMPORTANT | source-run and detector-response correlations           | CURRENTLY UNAVAILABLE |
| SAGE covariance             | IMPORTANT | source-run, isotope and counter correlations            | CURRENTLY UNAVAILABLE |

| Data Object                 | Class     | Required Content                                       | Current Status        |
|-----------------------------|-----------|--------------------------------------------------------|-----------------------|
| Cross-experiment covariance | IMPORTANT | shared cross-section/source-line uncertainty treatment | CURRENTLY UNAVAILABLE |
| External veto covariance    | IMPORTANT | KATRIN/IceCube/solar uncertainty mapping               | CURRENTLY UNAVAILABLE |

#### 4. Extraction-Level Data Requirements

| Requirement                              | Class     | Purpose                                  |
|------------------------------------------|-----------|------------------------------------------|
| BEST extraction-by-extraction table      | ESSENTIAL | full likelihood and time/zone validation |
| BEST counter efficiencies                | ESSENTIAL | detector/systematics separation          |
| BEST source activity calibration details | ESSENTIAL | source normalization nuisance            |
| BEST background-only periods             | IMPORTANT | background subtraction validation        |
| GALLEX exposure-level table              | IMPORTANT | cross-experiment transfer                |
| SAGE run-by-run source exposure          | IMPORTANT | isotope and source validation            |

#### 5. Metadata Requirements

| Metadata                                     | Class                    | Purpose                     |
|----------------------------------------------|--------------------------|-----------------------------|
| exposure timestamps                          | ESSENTIAL for anisotropy | sidereal/time tests         |
| detector orientation / azimuth               | ESSENTIAL for anisotropy | lab vs celestial separation |
| source position and extension                | ESSENTIAL                | geometry weighting          |
| environmental logs                           | IMPORTANT                | drift and background checks |
| detector geometry files                      | IMPORTANT                | M4 geometry asymmetry       |
| machine-readable public baseline predictions | IMPORTANT                | reproducibility             |

## 6. Allowed Placeholder Assumptions

Allowed only if clearly labeled exploratory:

- diagonal covariance approximation;
- block-diagonal covariance approximation;
- conservative envelope for unknown correlated systematics;
- summary-level rate ratios;
- approximate geometry from public reports;
- placeholder sterile baseline with explicit limitation.

These cannot support strong validation claims.

## 7. Forbidden Statistical Shortcuts

Forbidden:

- presenting diagonal-error chi-squared as final likelihood validation;
- ignoring correlated systematics;
- suppressing covariance limitations;
- using only anchor points;
- silently dropping outlier experiments;
- combining experiments with incompatible uncertainty conventions without disclosure;
- fitting tensor anisotropy without metadata;
- claiming precision from rounded summary tables.

## 8. Missing-Data Handling

Every validation report must include:

- list of missing data;
- effect of each missing input;
- whether the result is exploratory or validation-grade;
- sensitivity to placeholder assumptions;
- explicit statement of what claim is blocked.

## 9. Machine-Readable Dataset Requirements

Each dataset should eventually provide:

- source lines and intensities;
- source activity and uncertainty;
- detector geometry;
- exposure times;
- extraction runs;
- observed rates/counts;
- statistical uncertainties;
- systematic covariance;

- detector efficiency model;
- background model;
- metadata for orientation and environment.

## LSC External Veto Checklist

Status: Step 2 validation infrastructure

Purpose: define required external constraints before any strong LSC claim

### 1. Rule

External vetoes are hard constraints.

If an LSC parameter region improves Gallium residuals but violates an external veto, that region is not viable.

### 2. Veto Summary

| Veto                       | Must Check                                                                        | Would Weaken LSC                                  | Would Falsify / Reject Region                    | Current Status |
|----------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|----------------|
| KATRIN                     | whether response implies beta-spectrum energy distortion                          | required response close to KATRIN-sensitive scale | required universal distortion excluded by KATRIN | unresolved     |
| IceCube                    | whether anisotropy/tensor behavior conflicts with neutrino anisotropy constraints | qualitative analogy only                          | mapped tensor amplitude exceeds bounds           | unresolved     |
| SME / Lorentz-style bounds | whether fixed tensor implies Lorentz-violation-like coefficients                  | no mapping exists                                 | implied coefficients exceed published limits     | unresolved     |
| solar neutrino consistency | whether scalar response conflicts with solar rates                                | solar inconsistency under same response           | required parameters break solar constraints      | unresolved     |

| Veto                            | Must Check                                                          | Would Weaken LSC       | Would Falsify / Reject Region                   | Current Status |
|---------------------------------|---------------------------------------------------------------------|------------------------|-------------------------------------------------|----------------|
| standard oscillation fits       | whether survival/transition behavior conflicts with 3-flavor fits   | tension under transfer | required behavior incompatible with global fits | unresolved     |
| detector calibration literature | whether calibration/crossmatch section uncertainties explain signal | M1/M2/M4 LSC           | mundane explanation fully reproduces M6         | unresolved     |

### 3. KATRIN Checklist

Must check:

- whether LSC trace response implies universal energy-scale distortion;
- whether  $f0\_a(E, t)$  transfers to tritium beta-spectrum observables;
- whether response is detector-specific or universal;
- whether KATRIN constraints allow required amplitude.

Weakens LSC:

- required trace response is near a KATRIN-excluded scale.

Falsifies parameter region:

- same response required for Gallium would produce excluded KATRIN distortion.

Unresolved:

- no final frozen  $f0\_a$  mapping exists.

### 4. IceCube Checklist

Must check:

- whether anisotropy tensor is detector-local, lab-local, or celestial-frame;
- whether any claimed celestial tensor implies constraints from IceCube anisotropy or Lorentz-violation searches;
- whether IceCube is only a detector-systematics analogy or a quantitative veto.

Weakens LSC:

- IceCube is cited qualitatively without a calculation.

Falsifies parameter region:

- mapped anisotropy amplitude exceeds relevant IceCube bounds.

Unresolved:

- no machine-readable tensor-bound mapping is yet integrated.

## 5. SME / Lorentz-Style Tensor Bounds

Must check:

- whether fixed background tensor language implies SME-like coefficients;
- whether the detector-response tensor is purely detector-local or physical-background-like;
- whether published bounds constrain the required amplitude.

Weakens LSC:

- tensor is physically interpreted but no SME mapping is provided.

Falsifies parameter region:

- required tensor amplitude conflicts with SME/Lorentz bounds.

Unresolved:

- physical status of tensor remains intentionally limited to detector-response phenomenology.

## 6. Solar Neutrino Consistency

Must check:

- whether scalar response affects solar Gallium interpretation;
- whether the response is source-experiment-specific or universal;
- whether solar-neutrino rates remain consistent.

Weakens LSC:

- scalar response improves source experiments but worsens solar consistency.

Falsifies parameter region:

- required response breaks solar-neutrino constraints under the same assumptions.

Unresolved:

- response transfer rules are not frozen.

## 7. Standard Oscillation Fits

Must check:

- compatibility with three-flavor oscillation results;
- sterile-neutrino benchmark comparison under identical data treatment;
- whether LSC introduces effective behavior already excluded by global fits.

Weakens LSC:

- fit requires parameters inconsistent with established oscillation data.

Falsifies parameter region:

- required residual structure cannot coexist with global oscillation constraints.

Unresolved:

- no full global-fit integration exists.

## 8. Detector Calibration Literature

Must check:

- source activity uncertainties;
- gallium cross-section uncertainty;
- extraction efficiency uncertainty;
- background subtraction uncertainty;
- detector response/counter efficiency.

Weakens LSC:

- calibration/systematics can explain trace response.

Falsifies LSC preference:

- M4 or M2 reproduces M6-level structures after penalties.

Unresolved:

- M4 implementation is Step 2/Step 3 work.

## LSC Failure Modes And Risk Matrix

Status: Step 2 validation infrastructure

### 1. Risk Matrix

| Risk                        | Severity | Likelihood | Mitigation                                                      | Freeze Status  | Unresolved Dependencies                            |
|-----------------------------|----------|------------|-----------------------------------------------------------------|----------------|----------------------------------------------------|
| detector/system degeneracy  | Critical | High       | implement M4 and compare against M6                             | active blocker | detector calibration, extraction, background model |
| tensor underidentifiability | Critical | High       | predefine orientation, reduce free tensor use, require metadata | active blocker | timestamps, orientation, sufficient observables    |
| covariance absence          | Critical | High       | obtain full covariance or label exploratory                     | active blocker | BEST/GALLEX/SAGE covariance                        |
| overfitting                 | Critical | High       | LOO, cross-validation, penalties, model reduction               | active blocker | independent validation data                        |
| metadata incompleteness     | Critical | High       | require orientation/timestamps/environmental logs               | active blocker | BEST-2 metadata and release                        |
| external veto conflict      | High     | Medium     | quantitative KA-TRIN/IceCube/SME/solar checks                   | unresolved     | machine-readable constraints                       |
| parameter instability       | High     | Medium     | LOO stability and bounds reporting                              | unresolved     | covariance and transfer tests                      |



| Risk                         | Severity | Likelihood       | Mitigation                                           | Freeze Status  | Unresolved Dependencies             |
|------------------------------|----------|------------------|------------------------------------------------------|----------------|-------------------------------------|
| LOO instability              | High     | Already observed | reduce model complexity or reject current complexity | active warning | independent datasets                |
| false anisotropy detection   | Critical | High             | lab/celestial separation and M4 artifacts            | active blocker | orientation/timestamp metadata      |
| statistical overclaiming     | High     | High             | enforce covariance policy and claim boundary         | active blocker | review discipline                   |
| source normalization mimicry | High     | Medium           | independent source calibration nuisance              | unresolved     | source activity calibration records |
| cross-section mimicry        | High     | Medium           | nuclear cross-section uncertainty propagation        | unresolved     | literature and uncertainty model    |
| background drift             | High     | Medium           | background-only runs and environmental logs          | unresolved     | timestamped logs                    |
| selective baseline removal   | High     | Medium           | preregistration and mandatory M0-M6 reporting        | prohibited     | review enforcement                  |
| post-hoc tensor tuning       | Critical | Medium           | frozen orientation policy                            | prohibited     | preregistration                     |

## **2. Failure Modes**

### **FM1. M4 reproduces M6**

Result:

- LSC is not distinguishable.

Action:

- report as failure or unresolved depending on uncertainty;
- do not claim LSC support.

### **FM2. M1/M2 explain trace response**

Result:

- trace-only LSC is not supported.

Action:

- reduce claim to normalization/cross-section ambiguity.

### **FM3. LOO remains unstable**

Result:

- current complexity is overfit or underconstrained.

Action:

- reduce model complexity or defer validation.

### **FM4. Cross-experiment transfer fails**

Result:

- LSC lacks predictive generality.

Action:

- no cross-experiment support claim.

### **FM5. External veto conflict**

Result:

- affected parameter region is invalid.

Action:

- reject or revise only in a new preregistered version after human review.

### FM6. Metadata cannot identify anisotropy

Result:

- traceless/directional claim is untestable.

Action:

- freeze as unresolved; no anisotropy claim.

### FM7. Covariance remains unavailable

Result:

- likelihood-grade validation impossible.

Action:

- label all fits exploratory.

## 3. Risk Consolidation By Audit Source

| Audit Source           | Confirmed Risk                                                             |
|------------------------|----------------------------------------------------------------------------|
| Claude / physics audit | physical underderivation, tensor freedom, covariance, external vetoes      |
| Gemini                 | data limitations, covariance absence, orientation metadata absence         |
| DeepSeek               | detector/systematics mimicry, extraction and background artifacts          |
| Kimi                   | equation redundancy, trace/traceless freeze, LOO and validation discipline |

## 4. Freeze Status Summary

Frozen:

- risks must be reported;
- negative outcomes remain valid;
- M4 remains active;
- external vetoes remain hard constraints;
- covariance limitations block strong claims.

Not allowed:

- hiding risk;
- moving goalposts;
- expanding theory to avoid failure;
- using AI agreement as evidence.

# LSC M4 Detector/Systematics Baseline

Status: Step 2 validation infrastructure  
Role: required primary competitor to M6

## 1. Core Rule

M4 is not a nuisance footnote. M4 is a primary competitor to M6.

If M4 reproduces M6-level structures, LSC is not distinguishable.

## 2. Purpose

M4 tests whether ordinary detector, source, calibration, extraction, background or metadata effects can reproduce structures that LSC would otherwise describe as trace or traceless response.

M4 must be evaluated before any LSC preference claim.

## 3. M4 Components

| Component                          | Mechanism                                           | LSC Structure It Can Mimic         | Required Separation Test                                 |
|------------------------------------|-----------------------------------------------------|------------------------------------|----------------------------------------------------------|
| Calibration drift                  | time-dependent calibration shift                    | trace response or false time trend | calibration logs and stability runs                      |
| Source normalization uncertainty   | wrong source activity estimate                      | global deficit / trace term        | independent source activity calibration                  |
| Cross-section normalization        | gallium capture uncertainty                         | trace term                         | nuclear cross-section review and uncertainty propagation |
| Extraction efficiency asymmetry    | inner/outer or run-dependent extraction differences | zone structure / false anisotropy  | spiked recovery tests and zone-specific efficiency model |
| Detector geometry asymmetry        | hardware, dead volumes, guide tubes, tank structure | lab-frame anisotropy               | geometry model and rotation/source-position tests        |
| Background subtraction uncertainty | imperfect low-rate background model                 | trace or time-dependent residual   | background-only runs and timestamped event logs          |

| Component                    | Mechanism                                           | LSC Structure It Can Mimic | Required Separation Test                                   |
|------------------------------|-----------------------------------------------------|----------------------------|------------------------------------------------------------|
| Environmental drift          | temperature, radon, pressure, electronics           | false time/sidereal trend  | environmental sensor correlation analysis                  |
| Covariance incompleteness    | missing correlations                                | false significance         | full covariance or conservative correlated-error envelope  |
| Timing/orientation artifacts | lab-frame orientation confused with celestial frame | false traceless tensor     | orientation/timestamp metadata and sidereal-phase analysis |

#### 4. M4 Minimum Implementation Requirements

M4 must include:

- source activity nuisance;
- cross-section nuisance;
- zone-specific extraction efficiency nuisance;
- background subtraction nuisance;
- detector geometry asymmetry nuisance;
- environmental drift nuisance if time information exists;
- covariance-correlation nuisance;
- lab-frame orientation artifact term if anisotropy is tested.

These are not new LSC physics. They are detector/systematics competitors.

#### 5. M4 vs M6 Decision Logic

| Result                                                                    | Interpretation                |
|---------------------------------------------------------------------------|-------------------------------|
| M4 matches or exceeds M6 after penalties                                  | LSC not distinguishable       |
| M6 beats M4 but fails LOO                                                 | LSC unresolved / overfit risk |
| M6 beats M4 and passes LOO but lacks covariance                           | exploratory support only      |
| M6 beats M4, passes LOO, transfers across experiments and survives vetoes | candidate validation support  |
| M4 explains trace while M6 explains no additional structure               | trace-only LSC not supported  |

#### 6. Required M4 Outputs

Every M4 run must report:

- included systematics;
- excluded systematics and why;
- nuisance parameter count;
- covariance assumptions;
- fit quality;
- penalty-adjusted score;
- residual structure;
- comparison to M5 and M6;
- failure modes.

## 7. DeepSeek Detector-Attack Consolidation

DeepSeek-style detector audit conclusions are treated as mandatory constraints:

- cross-section normalization errors can mimic trace response;
- source activity miscalibration can mimic global deficit;
- extraction efficiency asymmetry can mimic zone structure;
- background/environment drift can mimic time or sidereal structure;
- missing orientation and timestamps make celestial anisotropy non-identifiable;
- incomplete covariance can inflate significance.

## 8. M4 Failure To Implement

If M4 is not implemented, then:

- M6 cannot be claimed preferred;
- directional/traceless claims remain unresolved;
- any fit result is exploratory only.

## LSC Preregistration Policy

Status: Step 2 validation infrastructure

Purpose: prevent post-hoc theory drift before BEST-2-era analysis

### 1. Rule

No BEST-2 validation claim is valid unless the analysis was preregistered before target data inspection.

### 2. Frozen Parameter Policy

Before fitting:

- list all active parameters;
- list all nuisance parameters;
- define parameter bounds;

- define which parameters are fixed;
- define which parameters are fitted;
- define penalty method;
- define failure behavior if parameters hit bounds.

Forbidden:

- adding parameters after fit failure;
- widening bounds after seeing data;
- hiding nuisance terms;
- treating tensor components as free without penalty.

### 3. Frozen Equation Policy

Before fitting:

- identify the active equation set from Step 1;
- identify any review-limited equations;
- exclude deprecated equation families;
- freeze all functional forms used in computation.

Forbidden:

- introducing new equations after data inspection;
- reviving PBH/curvature mechanisms;
- reclassifying historical equations as active because they improve a fit.

### 4. Frozen Orientation Policy

If anisotropy or sidereal behavior is tested:

- tensor orientation must be fixed before fitting;
- coordinate frame must be declared;
- lab-to-celestial transform inputs must be specified;
- detector orientation metadata must be included;
- environmental covariates must be handled.

Forbidden:

- selecting orientation after seeing residuals;
- fitting arbitrary tensor direction with insufficient data;
- claiming celestial anisotropy from lab-frame artifacts.

### 5. Anti-Posthoc Rules

Forbidden after data inspection:

- new mechanisms;
- new tensor structures;
- new particle claims;

- selective baseline removal;
- covariance suppression;
- unexplained data exclusion;
- changing success criteria;
- adding external-veto exceptions.

## 6. Blinded Analysis Rules

Recommended:

- freeze code and documents before unblinding;
- run M0-M6 on blinded or scrambled labels where possible;
- define all plots and tables before result inspection;
- archive negative and null outputs.

## 7. Allowed Tuning Boundaries

Allowed only before target data inspection:

- numerical implementation debugging;
- unit conversion fixes;
- documented data parsing correction;
- replacing placeholder covariance with official covariance;
- correcting transcription errors with provenance.

Not allowed:

- tuning response functions to match target residuals;
- changing tensor orientation after seeing residual phase;
- dropping M4 because it outperforms M6;
- changing external veto thresholds.

## 8. Publication Discipline

Every public release must state:

- validation grade: exploratory or validation-grade;
- which data were missing;
- which covariance assumptions were used;
- whether M4 was implemented;
- whether LOO passed;
- whether external vetoes were quantitative;
- whether negative outcomes occurred.

## 9. Negative-Result Acceptance

Negative outcomes are valid outcomes.

Examples:



- M4 beats M6;
- M1/M2 explain trace response;
- anisotropy unidentifiable;
- LOO unstable;
- external veto excludes required region;
- metadata insufficient for sidereal test.

These must be archived and cannot be rewritten as support.

## 10. Explicit Prohibitions

Prohibited:

- retrospective tuning;
- new mechanisms after fitting;
- selective baseline removal;
- covariance suppression;
- AI-consensus-as-evidence;
- theory expansion under Step 2;
- LSC 6.4 creation.

## LSC Validation Protocol

Status: Step 2 validation infrastructure

Scope: validation architecture only

Theory status: constrained, unvalidated phenomenological validation framework

### 1. Purpose

This protocol defines how LSC can be tested, constrained, weakened, falsified or rejected without extending the framework.

It does not:

- add mechanisms;
- add equations;
- create LSC 6.4;
- add tensor structures;
- add particle claims;
- reinterpret LSC as established physics.

### 2. Required Model Hierarchy

Every validation run must compare all active baselines:

| Model | Name                                    | Required Role                                  |
|-------|-----------------------------------------|------------------------------------------------|
| M0    | null / published baseline               | Reference expectation                          |
| M1    | scalar normalization only               | Tests whether deficit is simple scale shift    |
| M2    | cross-section / source-systematics only | Tests nuclear/source normalization explanation |
| M3    | sterile-neutrino benchmark              | Required physics competitor                    |
| M4    | detector-systematics benchmark          | Required primary mundane competitor            |
| M5    | LSC trace-only                          | Tests scalar LSC response                      |
| M6    | LSC trace + traceless anisotropy        | Tests full frozen LSC template                 |

M4 is a primary competitor to M6. If M4 reproduces M6-level structures, LSC is not distinguishable.

### 3. Mandatory Comparisons

Validation must report:

- M6 vs M0;
- M6 vs M1;
- M6 vs M2;
- M6 vs M3;
- M6 vs M4;
- M5 vs M1/M2;
- M6 vs M5;
- M4 vs M5/M6;
- cross-experiment transfer performance;
- leave-one-out stability.

No single comparison is sufficient.

### 4. Penalty Policy

All model comparisons must penalize complexity.

Allowed penalty classes:

- AIC-style information penalty;
- BIC-style information penalty;
- cross-validation penalty;
- preregistered nuisance-parameter penalty.

Forbidden:

- comparing raw fit quality only;
- counting tensor degrees of freedom as free but not penalized;
- adding nuisance terms without reporting them;
- fitting multiple variants and reporting only the best.

## 5. Covariance Policy

The preferred statistic is covariance-aware.

Requirements:

- use full statistical and systematic covariance when available;
- include correlated source-strength, cross-section, extraction and detector-efficiency terms;
- clearly label any diagonal or block-diagonal approximation as exploratory;
- never claim likelihood-level validation from diagonal-only errors when full covariance is missing.

If covariance is unavailable:

- validation status is limited to exploratory;
- no discovery, exclusion or strong preference claim is allowed;
- missing covariance must be listed as a blocker.

## 6. Leave-One-Out Requirements

LOO is mandatory because exact anchor fits are insufficient.

Required LOO modes:

- omit BEST inner, predict BEST inner;
- omit BEST outer, predict BEST outer;
- omit GALLEX combined/split, predict GALLEX;
- omit SAGE 51Cr, predict SAGE 51Cr;
- omit SAGE 37Ar, predict SAGE 37Ar;
- omit each experiment group, predict the omitted group.

Validation requires:

- residuals remain stable under omission;
- parameters do not swing into unrelated values;
- M6 does not degrade into a flexible interpolation of anchors;
- M4 and M3 are evaluated under the same omissions.

LOO instability is evidence against current model complexity.

## 7. Cross-Experiment Transfer Rules

Required transfers:

- fit BEST, predict GALLEX/SAGE;
- fit GALLEX/SAGE, predict BEST;
- fit  $^{51}\text{Cr}$  sources, predict  $^{37}\text{Ar}$  source behavior;
- fit inner/outer BEST split, test external gallium experiments.

Rules:

- transfer must use frozen mappings;
- no experiment-specific refit is allowed for a claimed prediction;
- failures must be reported, not absorbed into new nuisance terms.

## 8. External Veto Policy

External vetoes are hard constraints.

At minimum:

- KATRIN energy-response constraints;
- IceCube or SME-style tensor/anisotropy constraints where relevant;
- solar-neutrino consistency;
- standard three-flavor oscillation fits;
- detector calibration and cross-section literature.

If a parameter region improves Gallium residuals but violates a veto, it is not viable.

## 9. Detector/Systematics Separation Rules

Detector/systematics explanations must be modeled before LSC preference is claimed.

Required separation tests:

- source activity normalization;
- gallium cross-section normalization;
- inner/outer extraction efficiency asymmetry;
- geometry asymmetry;
- background subtraction drift;
- environmental drift;
- lab-frame orientation artifacts;
- covariance incompleteness.

If any of these reproduce LSC-like structures, LSC is unresolved or not distinguishable.

## 10. Preregistration Rules

Before BEST-2 fitting:

- freeze model equations from Step 1;

- freeze active parameters and bounds;
- freeze `f0_a` and `fA_a` if used;
- freeze orientation assumptions if anisotropy is claimed;
- freeze M0-M6 implementations;
- freeze success/failure criteria;
- freeze treatment of missing data;
- freeze external veto thresholds.

After data inspection:

- no new mechanism may be added under the same freeze;
- no selective baseline removal is allowed;
- no tensor reorientation is allowed;
- no covariance suppression is allowed.

## 11. What Counts As Validation

Validation requires all of:

- M5/M6 outperform simpler baselines after penalties;
- M6 outperforms M4 detector-systematics baseline;
- LOO remains stable;
- cross-experiment transfer succeeds;
- covariance-aware uncertainty is used or limitation explicitly blocks strong claims;
- external vetoes are survived quantitatively;
- negative and null outcomes are reported.

## 12. What Does Not Count As Validation

Not validation:

- exact anchor fits;
- visual agreement;
- diagonal-error chi-squared with missing covariance;
- AI audit agreement;
- fit improvement without complexity penalty;
- sterile-neutrino baseline omission;
- detector-systematics baseline omission;
- post-hoc tensor orientation;
- adding terms after seeing failures.

## 13. What Counts As Failure

Failure or major weakening occurs if:

- M4 reproduces M6-level structures;
- M1/M2 perform as well as M5/M6 after penalties;

- LOO remains unstable;
- cross-experiment transfer fails;
- anisotropy collapses into scalar normalization under covariance;
- external vetoes exclude required parameter regions;
- tensor parameters are unidentifiable.

## 14. What Counts As Unresolved Ambiguity

Unresolved:

- summary data support a deficit scale but cannot separate trace and traceless components;
- covariance is missing;
- orientation/timestamp metadata are missing;
- M4 is incomplete;
- external veto mapping is qualitative only;
- parameter degeneracy remains.